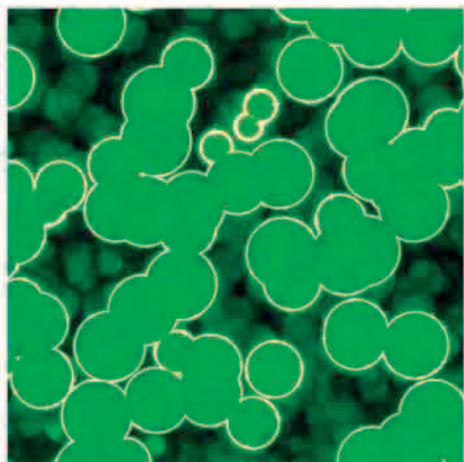


Kingdoms of life

Living things are divided into six kingdoms. Three of these are the animals, plants, and fungi—the relatively large, visible life forms that we notice all around us. The other three are the archaea, bacteria, and protists. These are microscopic organisms that are too small to see with the naked eye, yet they are vital to the web of life on Earth.

Bacteria and archaea



Number of known species 36,000

Habitat Wet places

Size range Microscopic to 0.03 in (0.75 mm)

Archaea and bacteria are simple, single-celled organisms. Though similar in many ways, they have different biochemistry, which shows that they have evolved separately over the billions of years since life first appeared. Because of this, they are now classified as separate kingdoms. Some of these organisms cause diseases, but many more are vital to the other four kingdoms of life.

Fungi

Number of known species 100,000

Habitat Land

Size range Microscopic to about 3.5 sq miles (9 sq km)

Fungi feed on organic matter such as dead plants. They are most familiar as the mushrooms and toadstools that spring up almost overnight in damp places, but these are only the visible parts of much larger root networks that spread over large areas. The largest known example is a honey fungus that covers almost 3.5 sq miles (9 sq km) of forest in Oregon. Other fungi, known as yeasts, are tiny, single-celled microbes.



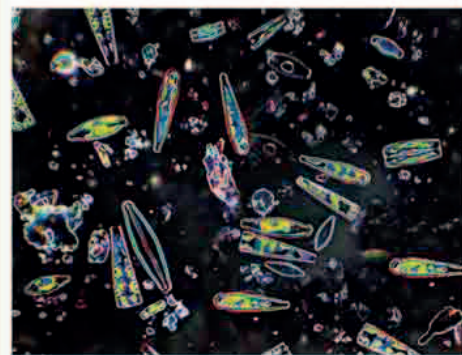
Protists

Number of known species 115,000

Habitat Water

Size range Microscopic to about 150 ft (45 m)

These are mostly single-celled organisms, but each cell has a nucleus—a central part containing genetic material. Some, such as the phytoplankton that drift in oceans and lakes, live like plants by using the energy of sunlight to make food. Others, such as *Paramecium*, behave like tiny animals. Many live in colonies, and a few, such as seaweed, are truly multicellular organisms.



▲ PHYTOPLANKTON

Tiny, single-celled algae drift in water, making food in the same way as plants.

Plants

Number of known species 310,000

Habitat Land and water

Size range Less than 0.04 in (1 mm) to about 280 ft (85 m)

These multicellular organisms nearly all use the energy of sunlight to turn carbon dioxide and water into sugar—a process called photosynthesis. They include primitive mosses and ferns, such as this one, and more complex plants, ranging from grasses to giant, long-lived trees.

